Introduction to Digital Musicology

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Course team

Anna Aljanaki

- Professor of Computational Methods in Time-Based Arts and Musicology
- Lecturer

Dominik Lekavski

- Third year master student in Computer Music
- Teaching assistant

Course organization

Lectures	02.03, 09.03, 23.03, 20.04, 27.04, 11.05
Practice Sessions	16.03, 13.04, 04.05, 18.05, 01.06
Seminars	08.06, 15.06, 22.06
Homeworks	There will be 5 homeworks related to each practice session
Tests	6 tests on the topic of each lecture, in moodle
Group project	The course ends with a group project implementing some digital musicological research

Criteria

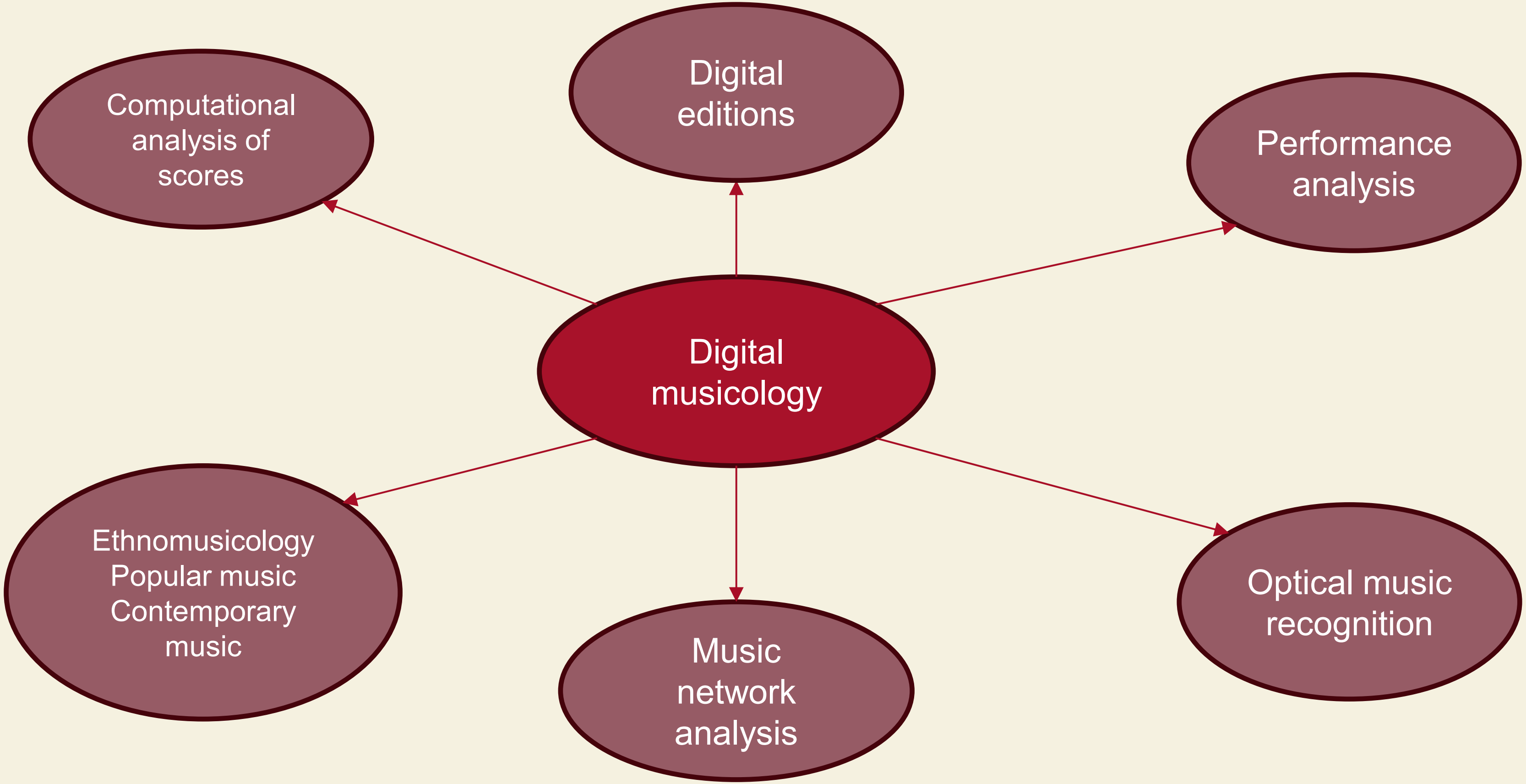
- Attend at least 4 out of 6 lectures and participate in discussions
- Homeworks constitute 30% of the grade (each worth 6 points)
- Tests are 24% of the grade (each worth 4 points)
- Project is 46% of the grade

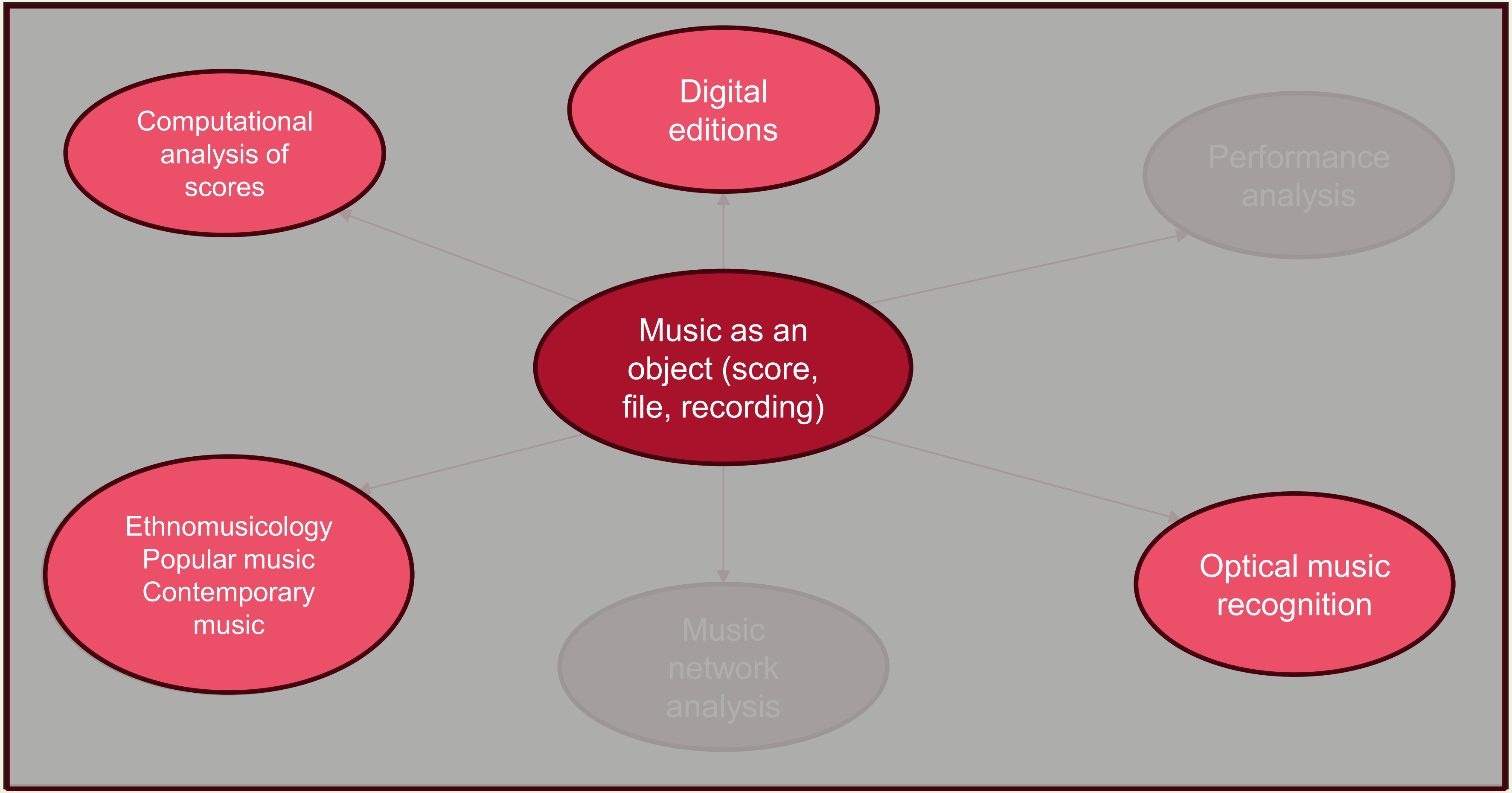
What is digital (computational) musicology?

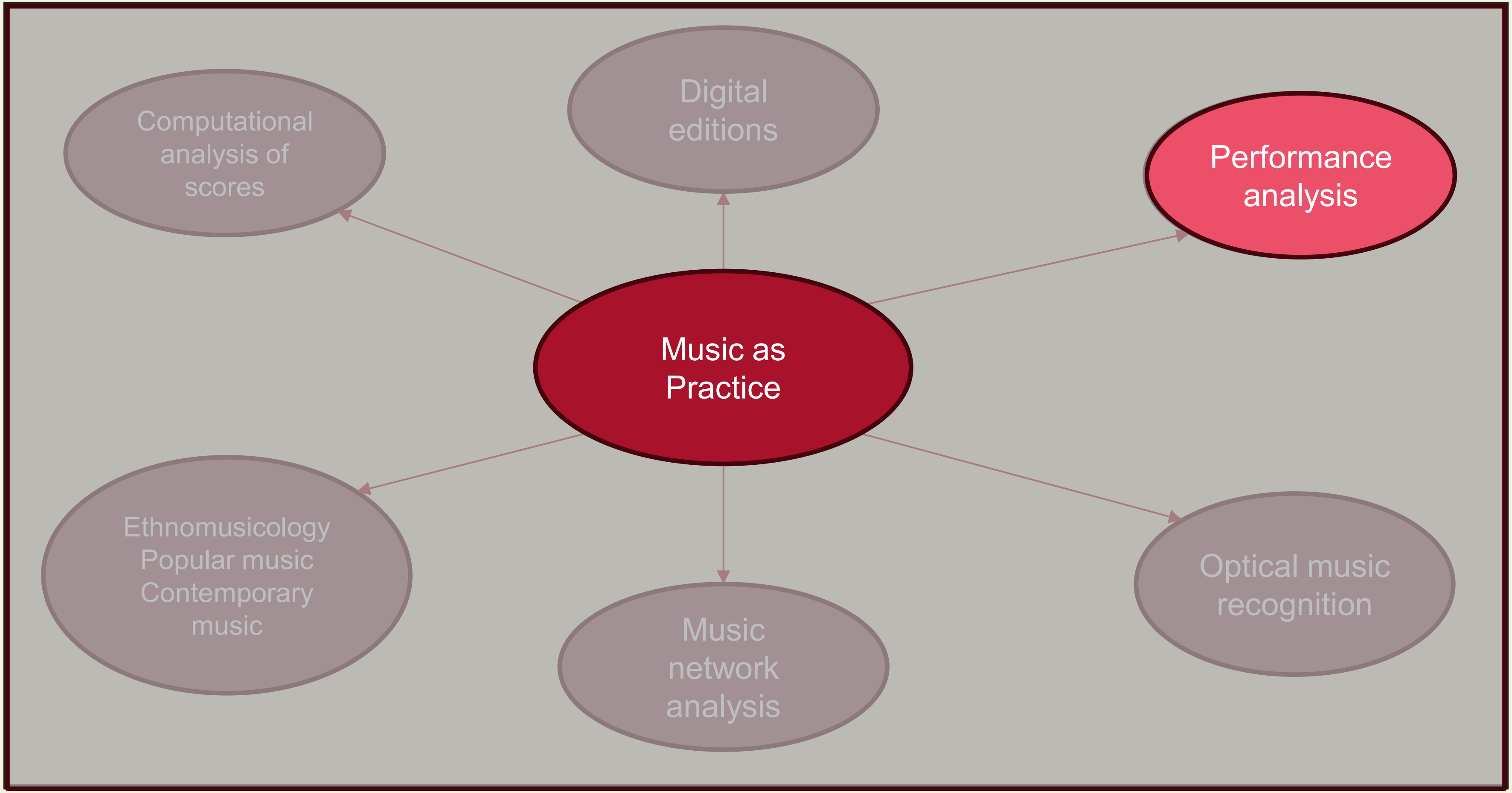
Examples, definitions and more

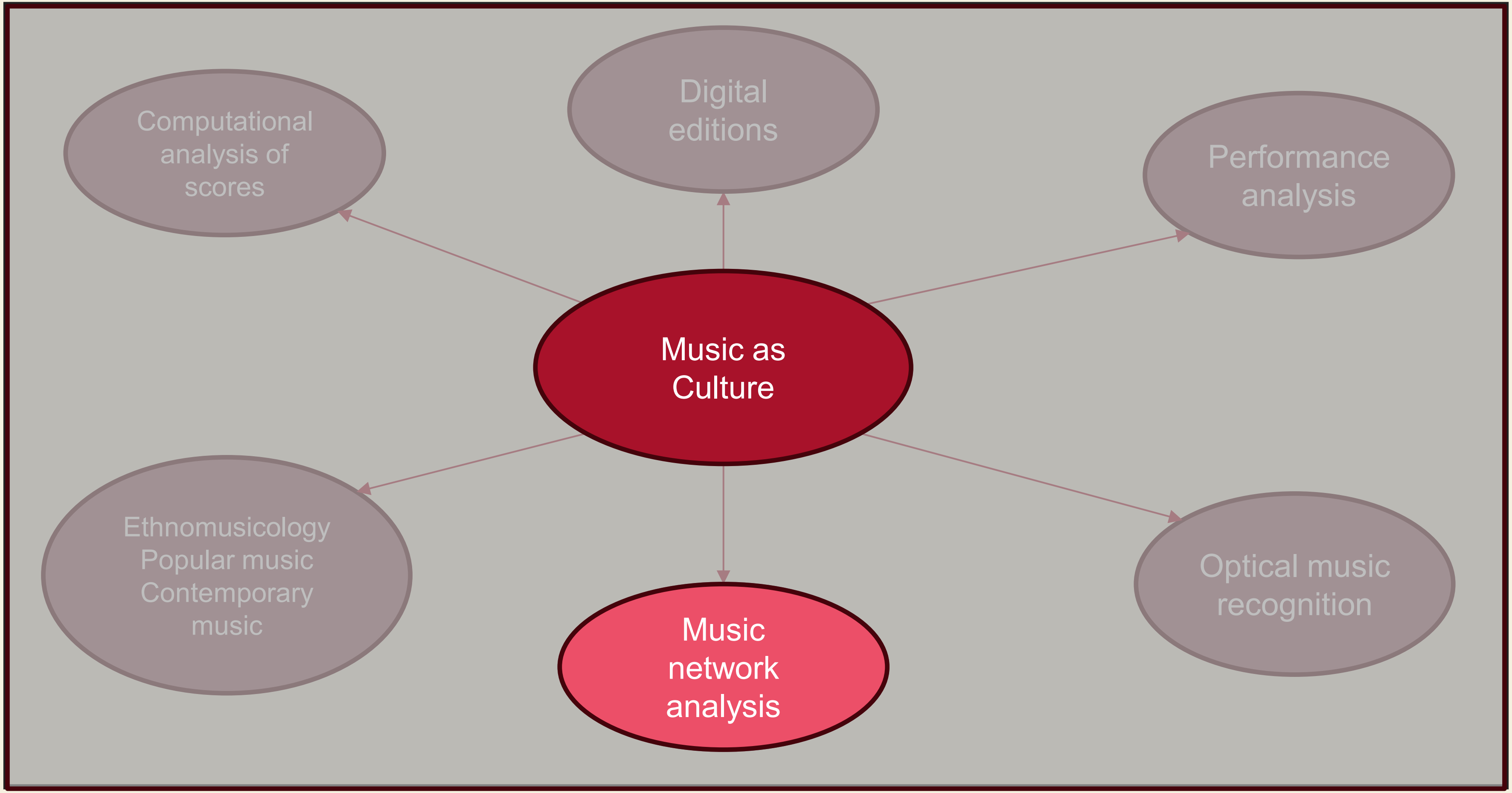
Why use computation for music?

- Enables working with large scale corpora
- Create digital editions
- Some of the musical data is naturally occurring in formats that are very suitable for computer processing


Digital musicology
Computational analysis of scores
Digital editions
Performance analysis
Ethnomusicology
Popular music
Contemporary music
Music network analysis
Optical music recognition


Computational analysis of scores
Digital editions
Performance analysis
Music as an object (score, file, recording)
Ethnomusicology
Popular music
Contemporary music
Music network analysis
Optical music recognition


Computational analysis of scores
Digital editions
Performance analysis
Music as Practice
Ethnomusicology
Popular music
Contemporary music
Music network analysis
Optical music recognition


Computational analysis of scores
Digital editions
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Music as Culture
Ethnomusicology Popular music Contemporary music
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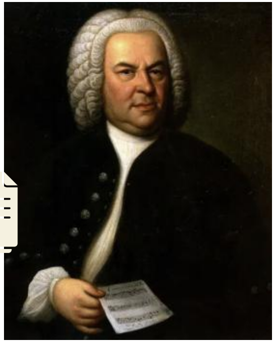
Case study

1. Authorship attribution: a case of 7 organ fugues



Johann Sebastian Bach
1685 - 1750

Wilhelm Friedemann Bach
(eldest son of Bach)
1710 - 1784

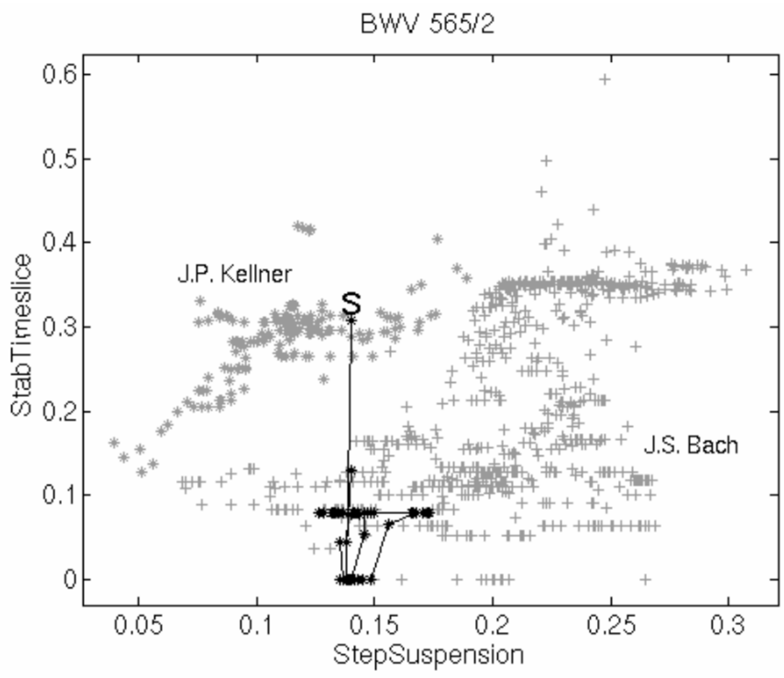
Johann Ludwig Krebs
(Bach's pupil)
1713 - 1780

Johann Peter Kellner
(Bach's admirer and distributor)
1705 - 1772

7 organ fugues BWV 534/2, 536/2, 537/2, 555/2, 557/2, 560/2, 565/2

Peter van Kranenburg (2008). On measuring musical style – the case of some disputed organ fugues in the J. S. Bach (BWV) catalogue. Computational Musicology, 15

Peter van Kranenburg, P. and Backer, E. (2005). Musical style recognition — a quantitative approach. In Handbook of Pattern Recognition and Computer Vision, pages 583–600



7 organ fugues BWV 534/2, 536/2, 537/2, 555/2, 557/2, 560/2, 565/2

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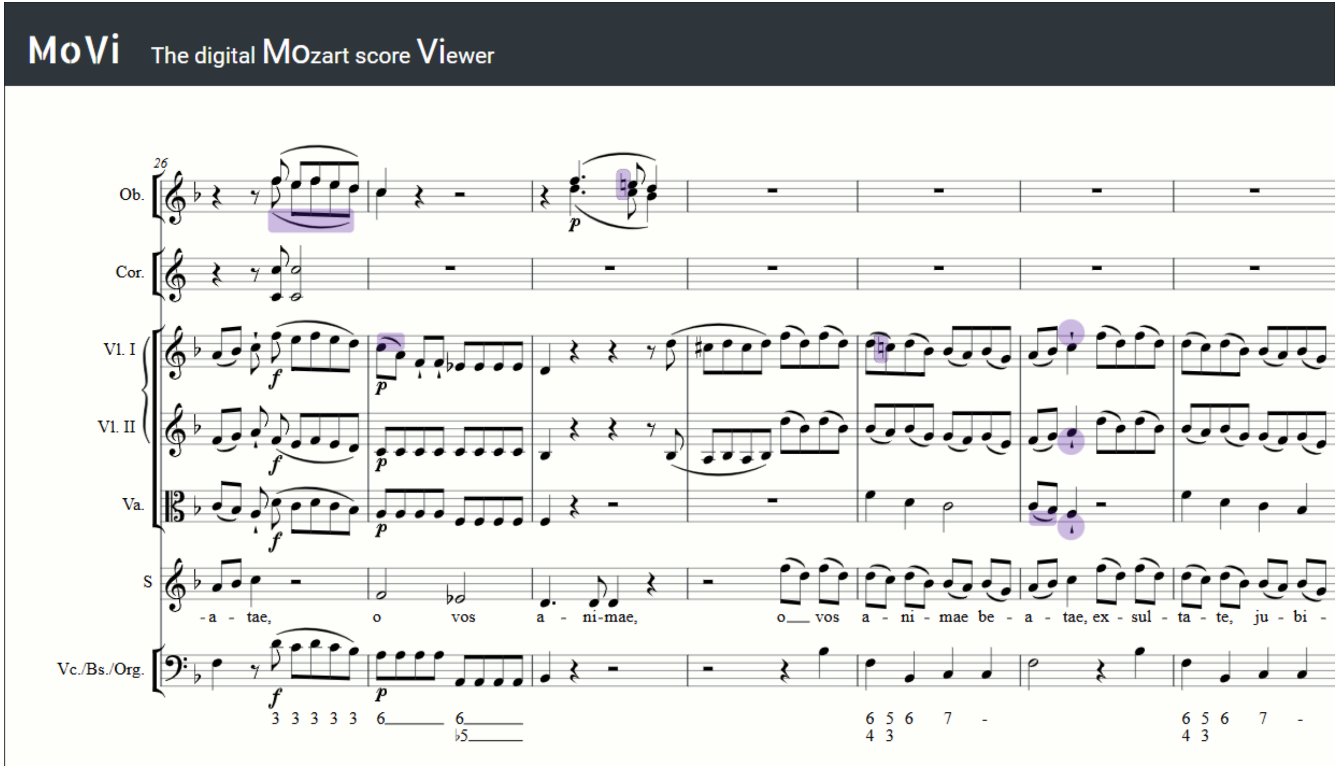
2. Digital editions

Digital Critical Editions

- integrate digitized (usually in MEI format) enriched source materials
- allows users to explore various stages of a work’s transmission, editorial interpretations

The case of BWV 903

- J. S. Bach’s Chromatic Fantasia and Fugue
- No surviging autograph
- 23 source manuscripts (attributed to Krebs, Bach’s pupils)
- Years 1717 (early draft, 1730 and later)



Mozarteum: https://dme.mozarteum.at/movi/en

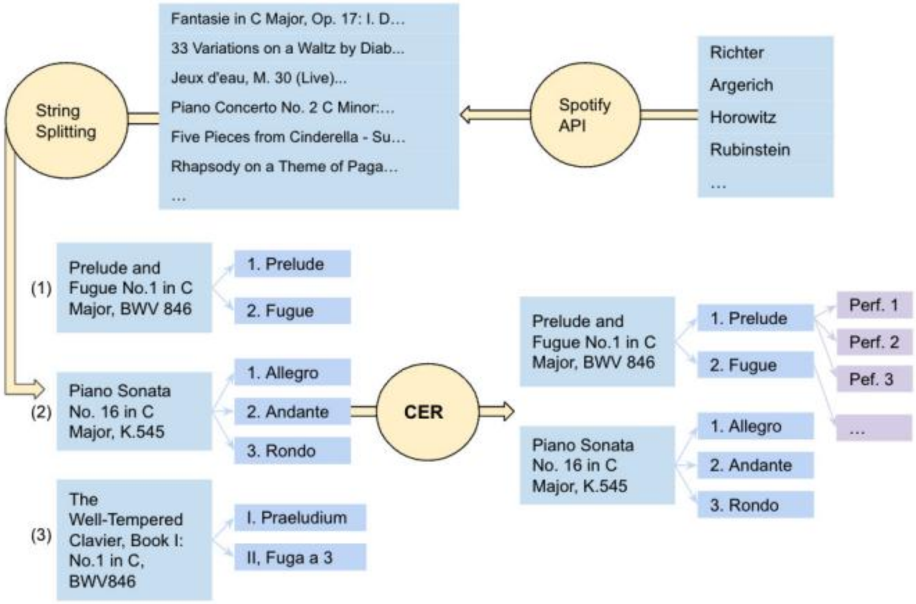
Other libraries

- Beethovens Werkstatt
  - https://beethovens-werkstatt.de/
- Manuskriptorium
  - manuscriptorium.com
- Center for Computer-Assisted Research in Humanities: Digital Resources for Musicology
  - https://drm.ccarh.org/
- Josquin Research Center
  - https://josquin.stanford.edu/

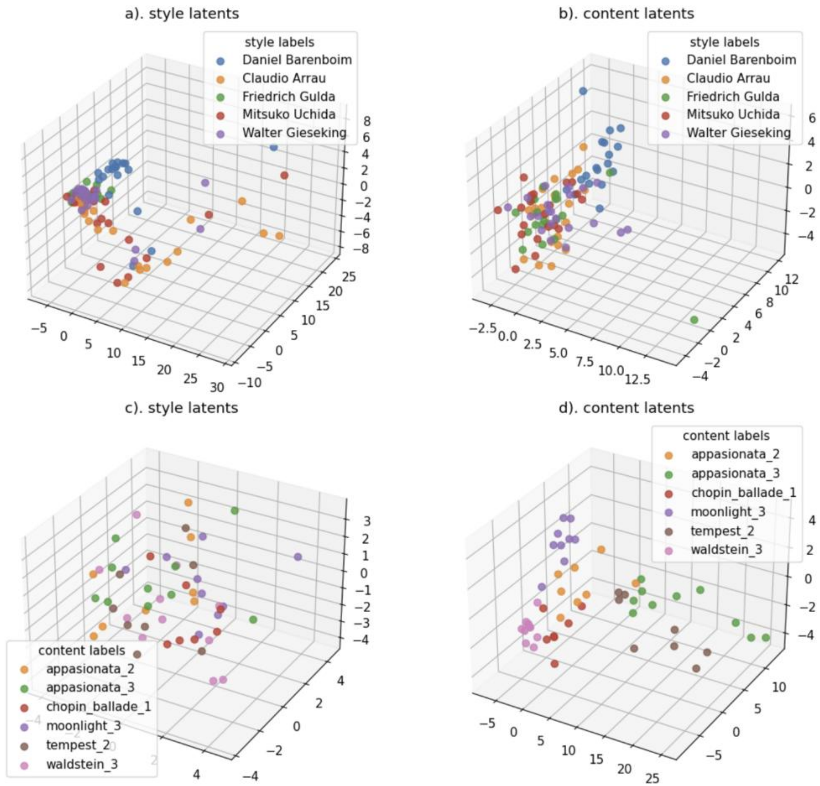
3. Performance study

Disklavier – the performance piano

Performance transcription - ATEPP

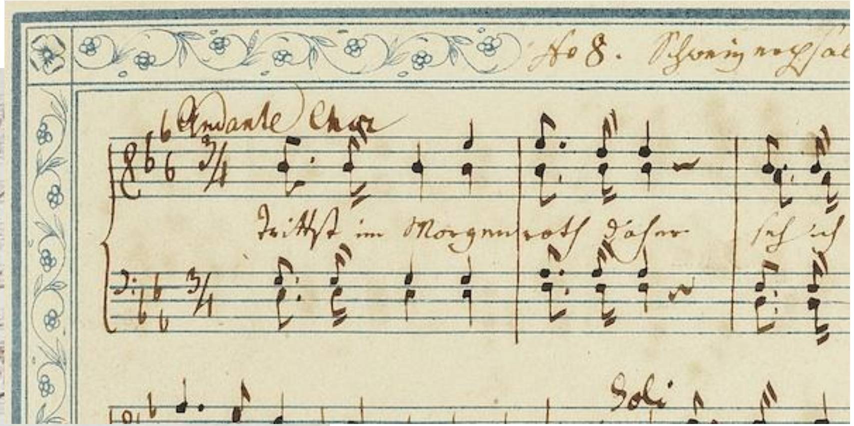


ATEPP: A Dataset of Automatically Transcribed Expressive Piano Performance. Zhang, H. et al., ISMIR 2022



H. Zhang and S. Dixon, "Disentangling the Horowitz Factor: Learning Content and Style From Expressive Piano Performance," *ICASSP 2023*

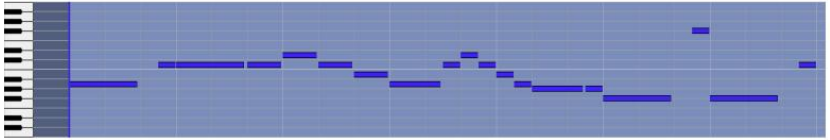
3.OMR



1.5 million records

Répertoire International des Sources Musicales: *https://rism.info/*

(a) Input: manuscript image.



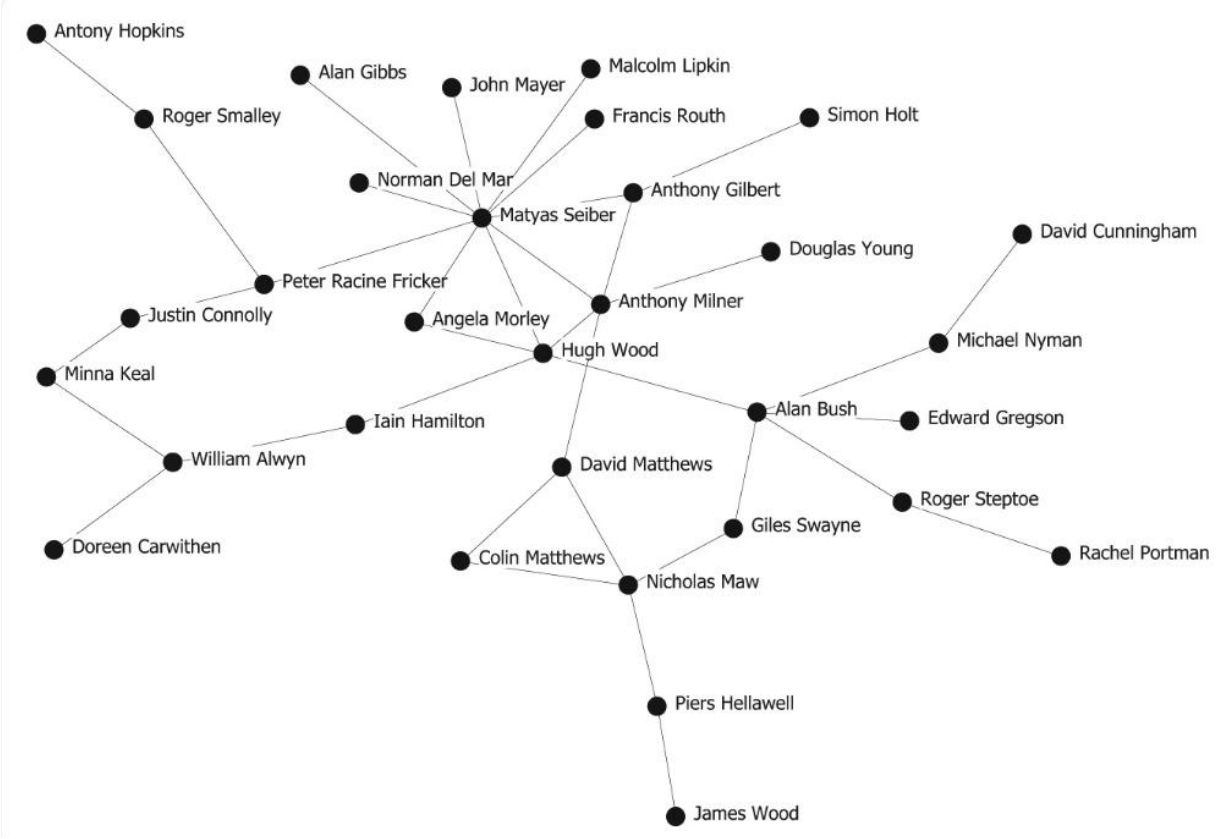
(b) Replayable output: pitches, durations, onsets. Time is the horizontal axis, pitch is the vertical axis. This visualization is called a *piano roll*.

(c) Reprintable output: re-typesetting.

(d) Reprintable output: same music expressed differently

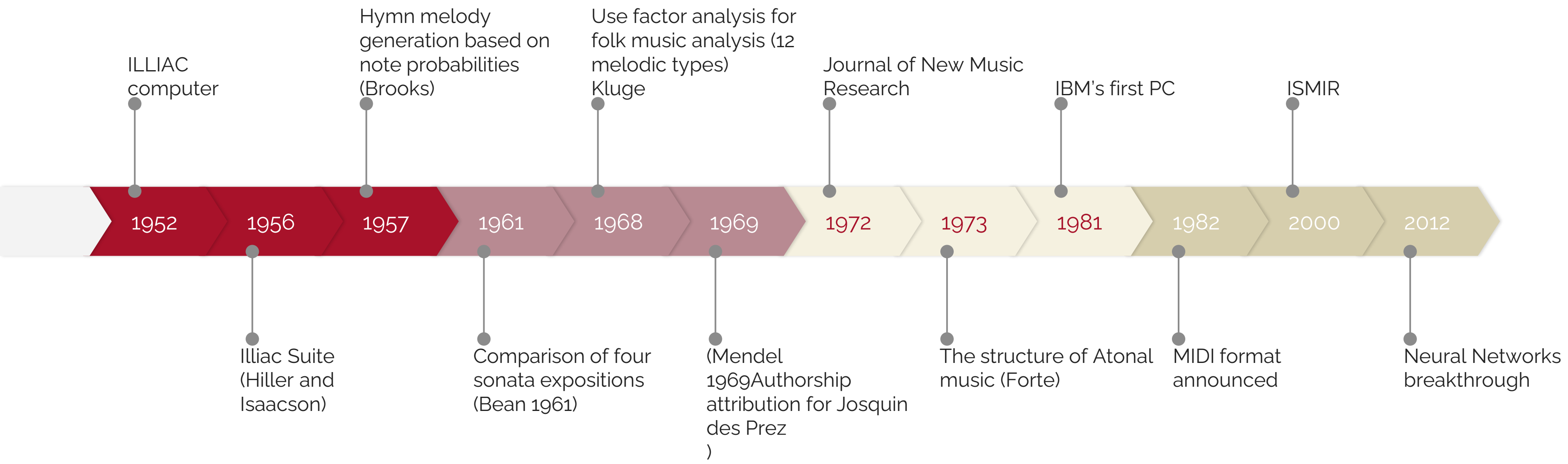
Hajic, J., & Pecina, P. (2017). In Search of a Dataset for Handwritten Optical Music Recognition: Introducing MUSCIMA++. *ArXiv, abs/1703.04824.*

4. Network analysis



McAndrew, S., & Everett, M. (2014). Music as Collective Invention: A Social Network Analysis of Composers. *Cultural Sociology*, *9*(1), 56–80

History of computational musicology



Schüler, N. (2005). Reflections on the History of Computer-Assisted Music Analysis I: Predecessors and the Beginnings. Musicological Annual, 41(1), 31-43.

Data structures and representations

What is data?

Formats of Data

Data Type	Examples
Images	Photos, scans, X-rays
Audio	Speech, music, bird song
Text	E-mails, books, articles
Time-series	EEG signal, stock prices
Spatial data	Coordinates, maps
Graphs	Social networks, airline routes
Numerical data	Age, height

Types of data

Data Type	Examples
Structured	Timetable, spreadsheet, gradebook
Unstructured	Emails, photos, interviews
Semi-structured	Server logs, webpages

Levels of measurement

Nominal	=, ≠	City of birth Gender Ethnicity
Ordinal	=, ≠, >, <	Top 5 Olympic medallists Language ability (e.g., beginner, intermediate, fluent)
Interval	=, ≠, >, <, +,−	Temperature in Celsius, IQ
Ratio	=, ≠, >, <, +,−, ×, ÷	Height, Age, Weight